

TSM 098 | INDEPENDENT PRACTICE

Polynomial operations and basic factoring

While TSM 099 studies Topic 17. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: like terms and distribution

1. Combine like terms.

$$(6x^2 + 2) + (-2x^2 - x - 2)$$

2. Combine like terms.

$$(5x^2 + 6x - 4) + (4x^2 + 2x + 5)$$

3. Combine like terms.

$$(-6x^2 - 5x - 2) + (5x^2 - 3)$$

4. Distribute and simplify.

$$5x(2x^2 + 4x - 2)$$

5. Distribute and simplify.

$$5x(x^2 + 4x - 5)$$

6. Distribute and simplify.

$$-3x(-x^2 - 3x - 5)$$

TSM 098 | INDEPENDENT PRACTICE

Polynomial operations and basic factoring

While TSM 099 studies Topic 17. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: polynomial multiplication

7. Multiply and combine like terms.

$$(x + 4)(2x - 2)$$

8. Multiply and combine like terms.

$$(x + 4)(x - 4)$$

9. Multiply and combine like terms.

$$(x - 1)(x + 3)$$

10. Multiply and combine like terms.

$$(x + 4)(x + 3)$$

11. Multiply and combine like terms.

$$(x^2 + x + 2)(x^2 + 2x + 3)$$

12. Multiply and combine like terms.

$$(x^2 + x - 3)(x^2 - x + 1)$$

TSM 098 | INDEPENDENT PRACTICE

Polynomial operations and basic factoring

While TSM 099 studies Topic 17. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: introductory factoring

13. Factor completely; check by multiplying.

$$x^2 + 7x + 12$$

14. Factor completely; check by multiplying.

$$x^2 - 8x + 12$$

15. Factor completely; check by multiplying.

$$x^2 + 2x - 15$$

16. Factor completely; check by multiplying.

$$x^2 - 6x - 16$$

17. Factor completely; check by multiplying.

$$6x^2 + 12x$$

18. Factor completely; check by multiplying.

$$x^2 - 16$$

TSM 098 | INDEPENDENT PRACTICE

Polynomial operations and basic factoring

While TSM 099 studies Topic 17. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: mixed polynomial practice

19. Multiply and simplify.

$$(x^2 + 2x + 1)(x^2 + x - 2)$$

20. Multiply and simplify.

$$(2x^2 + x - 3)(x^2 - x + 2)$$

21. Multiply and simplify.

$$(x + 2)(x^2 - 4)$$

22. Factor completely and verify.

$$x^2 - 3x - 28$$

23. Factor completely and verify.

$$x^2 - 10x + 25$$

24. Explain the error: only the first and last terms were multiplied.

$$(x + 2)(x + 3) = x^2 + 6$$